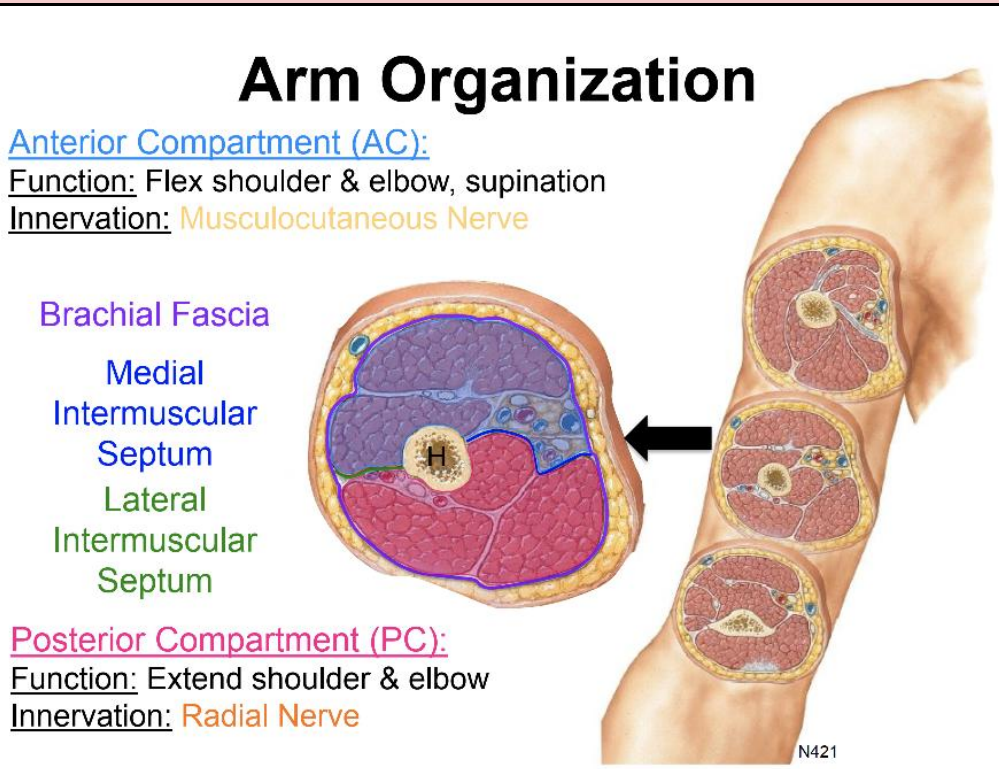
Week #8: [Muscles of the Arm]

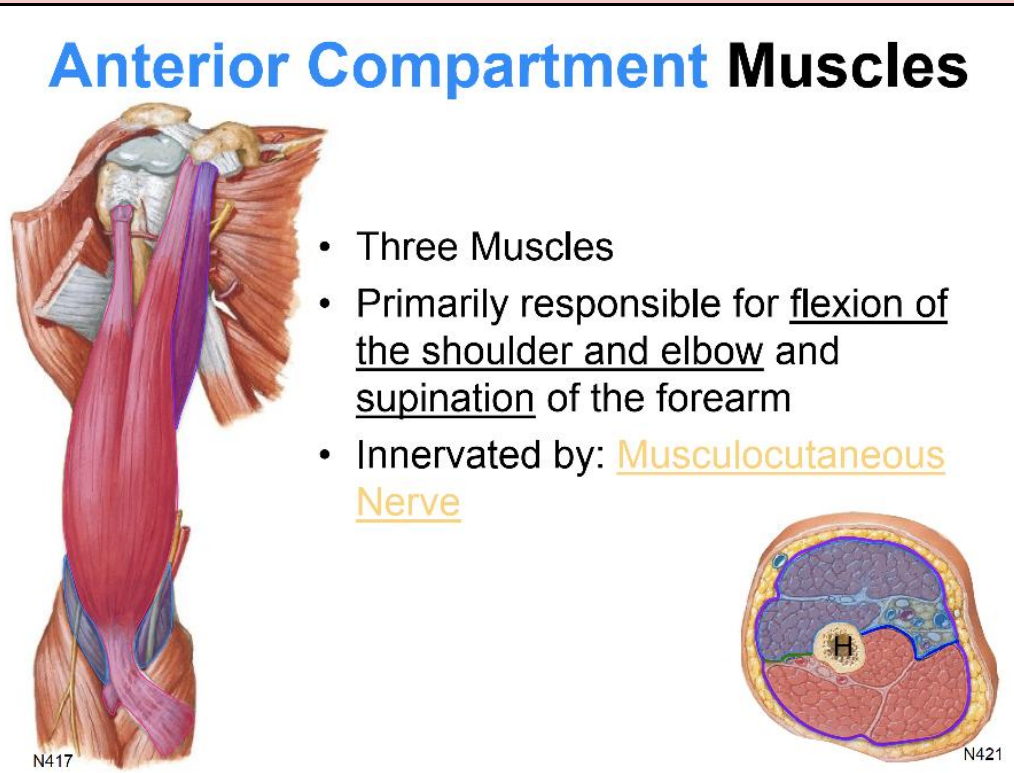
Clinical Approach to Muscles of the Proximal Upper Extremity

Approach to Proximal Biceps Brachii Tendinopathy **(Dr. Cassidy)**

Clinical Approach to Muscles of the Proximal Upper Extremity

- Muscles of the Arm





Arm Organization: The arm extends from the shoulder to the elbow. Deep to the skin and subcutaneous tissue is the deep fascia of the arm, known as **brachial fascia.** Two intermuscular septa (***medial and lateral intermuscular septa***) extend from the brachial fascia to the humerus to divide the arm into two compartments:

- Anterior compartment
- Posterior compartment

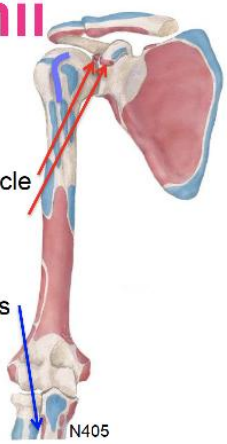
These compartments divide the arm into functional and innervation compartments

AC Muscles: Biceps Brachii

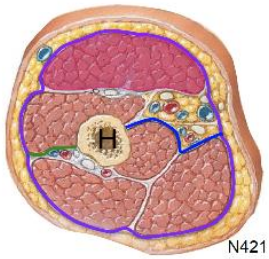


Origin:
Long head: Supraglenoid Tubercle
Short head: Coracoid Process

Insertion:
Radial Tuberosity & Aponeurosis



- Function:
  - Strong supinator
  - Flexes forearm
  - Weak shoulder flexor
  - Resists dislocation

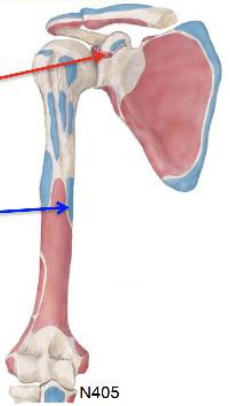


AC Muscles: Coracobrachialis

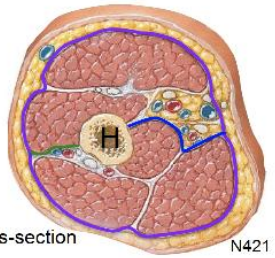


Origin:
Coracoid Process

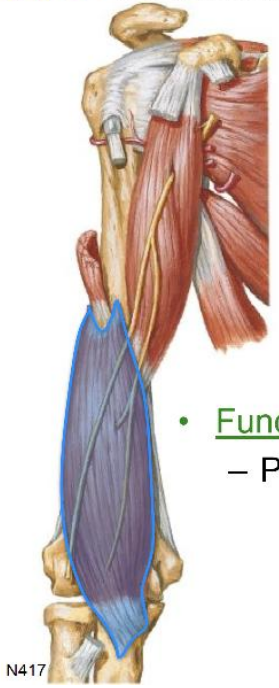
Insertion:
Mid ⅓ of Medial Humerus



- Function:
  - Flexes arm
  - Adducts arm
  - Resists dislocation



AC Muscles: Brachialis

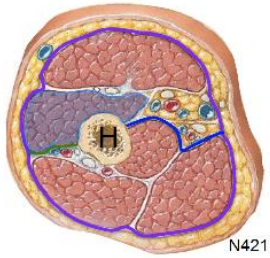


Origin:
Distal ½ of Anterior Humerus

Insertion:
Coronoid Process of Ulna



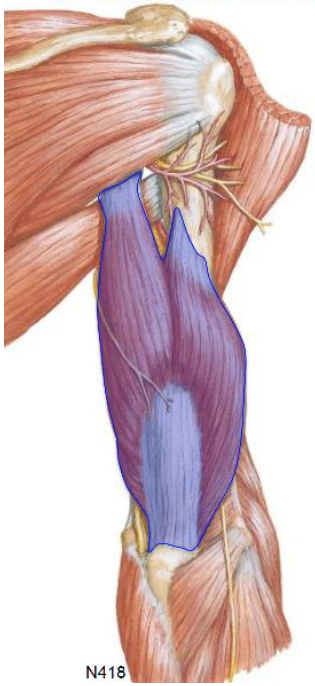
- Function:
  - Primary flexor of the forearm



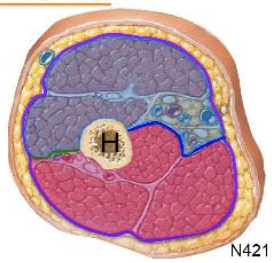
Anterior Compartment Muscles: function to ***flex the shoulder and elbow*** as well as ***supinate*** the forearm and they are all innervated by ***musculocutaneous nerve.***

Muscle	Origin	Insertion	Innervation	Function
Biceps Brachii	LH: Supraglenoid tubercle of scapula SH: Coracoid process	Radial tuberosity and forearm via bicipital aponeurosis	Musculocutaneous Nerve	Supinate & flex forearm; flex shoulder; resist dislocation
Coracobrachialis	Coracoid process	Mid 1/3 medial humerus		Flex and adduct shoulder; resist dislocation
Brachialis	Distal ½ anterior humerus	Coronoid process of ulna		Primary elbow flexor

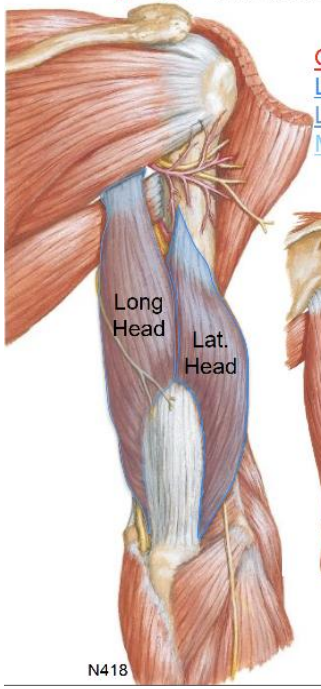
Posterior Compartment Muscle



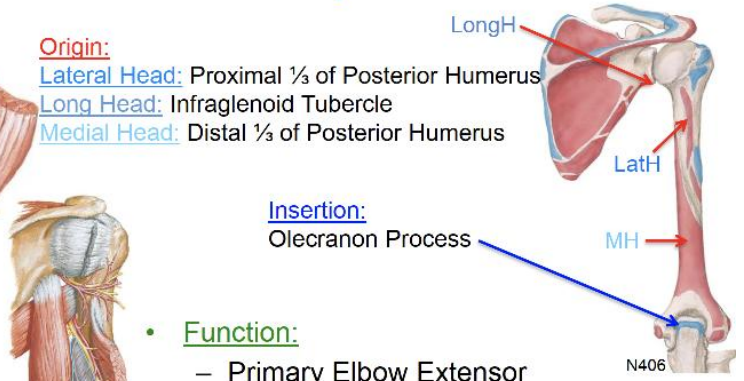
- One Muscle: Triceps
  - Three Heads:
    - Long Head
    - Lateral Head
    - Medial Head
- Primarily responsible for extension of the shoulder and elbow
- Innervated by: Radial Nerve



PC Muscle: Triceps Brachii



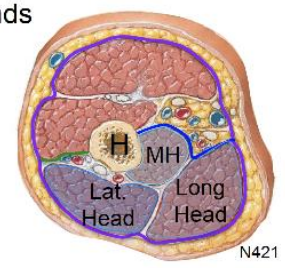
Origin:
Lateral Head: Proximal ⅓ of Posterior Humerus
Long Head: Infraglenoid Tubercle
Medial Head: Distal ⅓ of Posterior Humerus





Insertion:
Olecranon Process

- Function:
  - Primary Elbow Extensor
  - Long Head: extends shoulder joint & resists shoulder dislocation



Posterior Compartment Muscles: function to e***xtend the shoulder and elbow*** and they are all innervated by ***radial nerve.***

Muscle	Origin	Insertion	Innervation	Function
Triceps Brachii	Lg.H: Infraglenoid tubercle of scapula Lat.H: Prox. 1/3 Post. Humerus Med.H: Distal 1/3 Post. Humerus	Olecranon process of ulna via common tendon	Radial Nerve	Primary extensor of the elbow joint; Long head – extends shoulder and resists dislocation at shoulder joint

Knowledge Check

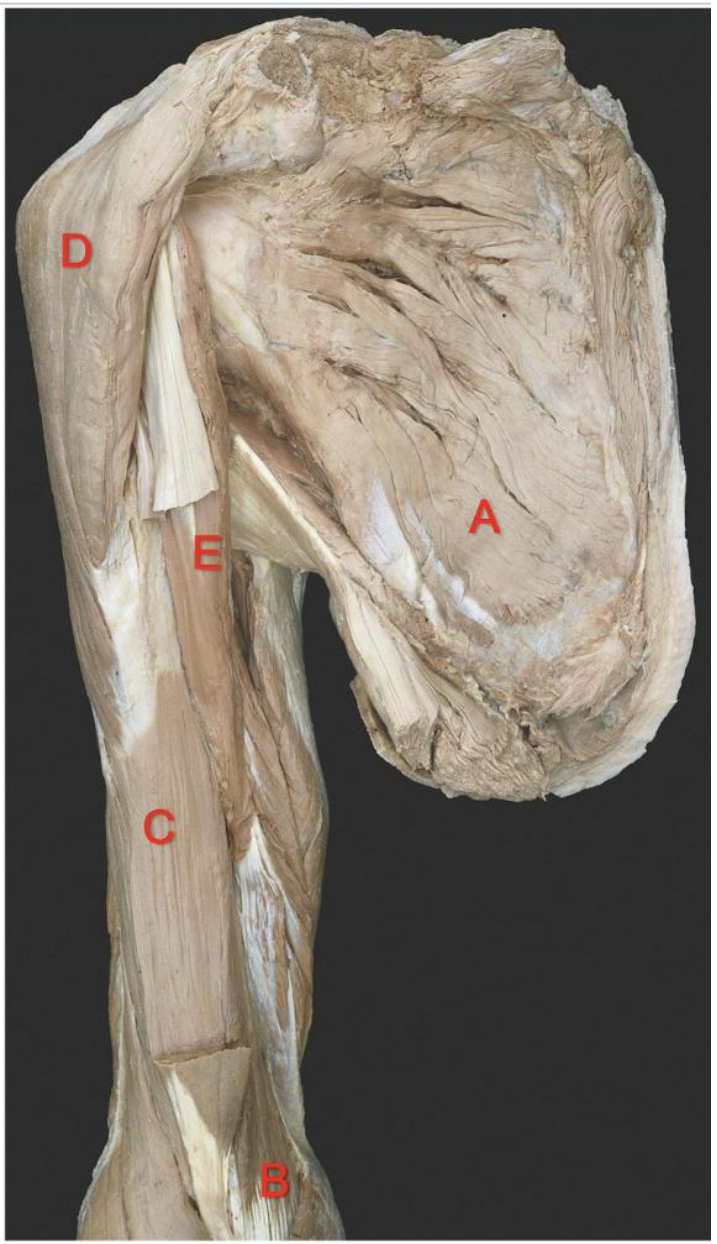
Biceps Brachii C

Brachialis B

Coracobrachialis E

Deltoid D

Subscapularis A



Approach to Proximal Biceps Brachii Tendinopathy (Dr. Cassidy)